

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2019- 2020 (Odd Semester)

Innovative Practices Description

UNIT I Classification of Signals and Systems

Degree, Semester& Branch: III Semester B.E ECE A

Course Code & Title: EC8352 Signals and Systems

Name of the Faculty member: Mrs.R.Chitra

Name of the topic: Classification of Systems

Name of the Activity: One Minute Paper

Date & Duration: 12.07.2019 & 15 minutes

Description:

Minute paper is a short writing activity within the classroom in which students generate answers in response to questions asked by the teacher regarding the material taught in class. While the original purpose of the one-minute paper was to assess student learning at the end of a day's lesson. Students must first organize their thinking to rank the major points and then decide upon a significant question.

Goals (Learning Outcomes):

1. The students will be able to learn the fundamental concepts in classification of systems.
2. The students can be acquired better knowledge in classification of systems.

Use of appropriate methods:

Justification for choosing the topic using One Minute Paper activity:

There are many classifications of systems based on parameter used to classify them. As it may be confused, making it as a practice to write those equations is a healthy aspect to remind those equations. As an exam point of view, it is frequently asked topic in University exam.

Use of appropriate methods:

Justification for choosing the topic using One Minute Paper activity:

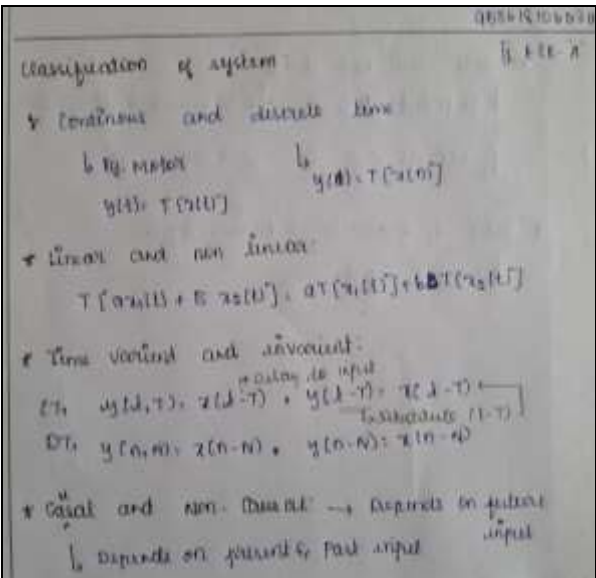
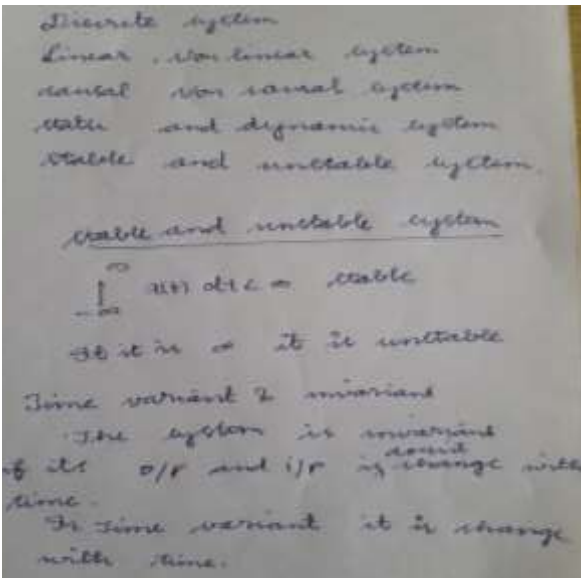
Link budget is an important topic that involves many equations such as carrier to noise ratio, EIRP, Flux density, Back-off etc. As it may be confused, making it as a practice to write those equations is a healthy aspect to remind those equations. As an exam point of view, it is frequently asked topic in University exam.

Effective presentation

Implementation (Plan & Execution) with Proof:

- ✓ Minute papers are an effective way of involving all students in class simultaneously. It ensures equal participation of each and every class member, including anyone who may be too shy or fearful to participate orally.
- ✓ Each and every student is expected to participate and has something important to contribute - no matter what their cultural background or prior level of academic preparedness

Sample Images:

UNIT I			
1	12.07.19	Classification of systems	One Minute Paper
 			

Significance of Results:

Assessment of Effectiveness/Success of the Activity:

- By conducting this activity, the students were able to recall the topics and they were able to score good marks for the questions under this topic during Internal Assessment Test I.
- They were able to realize the point of small mistakes that they may commit while using the properties of the system.

Reflective Critique:

• Benefits:

- Minute papers can improve the quality of class discussion by having students write briefly about a concept or issue before they begin discussing it.
- I have found that this gives the more reflective chance to the students to gather their thoughts prior to verbalizing them and benefits that student who are more fearful of public speaking by giving them a script to fall back on (or build on)

• Challenges:

- As it is mentioned above that it takes one minute to conduct this activity. But, the students are failed to remember all equations in a given duration. They took five minutes to complete this activity.

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2	2	1					2	1		

CO1: The students will be able to analyze the various properties of signals and systems.

References:

1. Poorna Chandra S, Sasikala. B.Digital Signal Processing, Vijay Nicole/TMH, 2013.
2. <https://provost.tufts.edu/celt/files/MinutePaper.pdf>

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Innovative Practices Description

UNIT II Analysis of Continuous Time Signals

Degree, Semester & Branch: III Semester B.E ECE A

Course Code & Title: EC8352 Signals and Systems

Name of the Faculty member: Mrs.R.Chitra

Name of the Topic: Fourier Series-Properties

Name of the Innovative Practice: Zero Minute Speech

Date & Duration: 26.07.19 & 10 minutes

Description:

Zero Minute Speech will lead to express basics of the particular topics by different students. The important topics were allotted to set of students and they will give speech on the topic with doubt clarification.

Goals (Learning Outcomes):

- The students will be able to represent continuous time periodic signals using Fourier series.

Use of appropriate methods:

Justification for choosing the topic using Zero Minute Speech activity:

Usually the students are not interested in studying the properties of any transform and not attending properties related questions in the examination. Therefore, I have chosen this topic to change student predetermined mindset and to make students to attend questions under this topic. Also it leads to express basics of the particular topics by different students.

Effective presentation

Details of the Implementation:

- Initially I asked all students to recall the contents learnt in the class room for ten minutes
- Later I called students to talk about the properties of Fourier series. But, students are felt shy to come front and talk.
- For motivating the students, I announced that I will give 2 bonus marks for assignment submission. Then students got motivated and started to come and represent their views about this topic.

Sample Images



Reflective Critique:

Benefits:

- Few students share their ideas and information with classmates and some students interacted with other students to clarify the doubts.
- Communication skills can be improved
- Students may feel free to come and talk in front of their class mates

Challenges:

- Not able to give chance to all students during zero minute speech practice.
- Few students hesitate to come to the Dias and share the concept.
- Consumed more time than what I planned

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO2	3	2	2	1	1			1		2		2

CO2: The students will be able to apply Laplace transform and Fourier transform in signal analysis.

References:

1. <https://www.khanacademy.org/science/electrical-engineering/ee-signals/ee-fourier-series>
2. <http://www.icaltefl.com/just-a-minute-speaking-activity>

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Innovative Practices Description

UNIT III Linear Time Invariant – Continuous Time Systems

Degree, Semester & Branch: III Semester B.E ECE A

Course Code & Title: EC8352 Signals and Systems

Name of the Faculty member: Mrs.R.Chitra

Name of the Topic: Impulse response, Convolution Integral

Name of the Innovative Practice: Quiz Dominos

Date & Duration: 19.08.19 & 15 minutes

Description:

Dominoes – key words/terms and their definitions are written on cards given to the group of students. A key word/term is read out and the one person with the correct definition for this key word/term from some other group stands up and reads out their definition and so on... this continues in a loop until you are back to the start – great fun when you put learners against the clock.

Goals (Learning Outcomes):

- The students will be able to apply the properties of impulse response and convolution integral to analyze Linear Time Invariant CT Systems.

Use of appropriate methods:

Justification for choosing the topic using Quiz Dominos activity:

A convolution is a mathematical way of combining two signals to form a third signal which is used to analyze the LTI (Linear and Time-Invariant) system or filter. Students must know about this topic. But, it is difficult to study for students. To make the students feel easy with this topic, I conducted this activity (Quiz Dominos) for this topic which made the students to understand better.

Effective presentation:

Details of the Implementation:

- I have prepared some keywords/questions related to convolution integral and its properties and wrote one key word/question per chart paper (small).
- Made all students to sit in a team(3-4 students)
- Distributed one chart paper per team
- Initially I asked one team to read given key word in the given chart paper
- The one student with the correct definition for this key word/term from some other team stands up and reads out their definition and so on... this continues in a loop until all questions are read and find out its answer

Sample Images



Reflective Critique:

Benefits:

- Students were able to understand easily
- Made entire class to be fully engaged during execution of the activity
- Can be improved their presence of mind
- Had a great fun when i put learners against the clock.

Challenges:

- Planned 15 minutes but it took 20 minutes to complete this activity

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO3	2	3	2	1	1			1		2		2

CO3: The students will be able to analyze linear time invariant continuous time systems using Laplace and Fourier Transforms.

References:

1. <https://ocw.mit.edu/resources/res-6-007-signals-and-systems-spring-2011/video-lectures/lecture-4-convolution/>
2. <https://handsonaswegrow.com/dominoes/>

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Innovative Practices Description

UNIT IV Analysis of Discrete Time Signals

Degree, Semester & Branch: III Semester B.E ECE A

Course Code & Title: EC8352 Signals and Systems

Name of the Faculty member: Mrs.R.Chitra

Name of the Topic: Z transform

Name of the Innovative Practice: Think-Pair-Share

Date & Duration: 19.09.19 & 15 minutes

Description:

Think-pair-share (TPS) is a collaborative learning strategy where students work together to solve a problem or answer a question about an assigned reading. This strategy requires students to (1) think individually about a topic or answer to a question; and (2) share ideas with classmates. Discussing with a partner maximizes participation, focuses attention and engages students in comprehending the reading material.

Goals (Learning Outcomes):

- The students will be able to analyze discrete time signals using Z transform.

Use of appropriate methods:

Justification for choosing the topic using Think-Pair-Share activity:

The Z Transform has a strong relationship to the DTFT, and is incredibly useful in transforming, analyzing, and manipulating discrete calculus equations in the field of digital signal processing. It will be useful to study Digital Signal Processing subject in V Semester to analyze digital filters and to analyze the linear discrete system. So, I have chosen this topic to conduct this activity for making students to share their ideas among their peers.

Effective presentation

Details of the Implementation:

- I have planned this activity for 15 minutes
- Initially I have explained what students need to try and asked the students to form a pair
- Made all the students to engage in participation
- Students were discussed about the given topic with each other in talk partners for five minutes.
- After that, they are asked to express their thoughts to an audience and contribute to a shared solution.
- Make sure there is enough time for both students in a pair to discuss their ideas.

Sample Images:



Reflective Critique:

Benefits:

- This step ensures that every student has the opportunity to explain their thoughts to others and listen to others conversations.
- Due to this activity, even though one student leaves any point the other student will help to recollect the ideas. So they felt comfortable to think back and share it.
- Students were able to remember the content of the topic by conducting this activity and maximum students were attended the questions under this topic in Internal assessment Test.

Challenges:

- One of the biggest challenges of the think-pair-share is to get all students to truly be engaged
- Most pairs shared the same ideas about the topic
- It took more time than what I planned

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO4	2	3	2	1	1			1		2		2

CO4: The students will be able to analyze discrete time signals using DTFT and Z transform.

References:

3. Poorna Chandra S, Sasikala. B.Digital Signal Processing, Vijay Nicole/TMH, 2013.
2. www.et.iitb.ac.in › Resources › files › TPS

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Innovative Practices Description

UNIT V Linear Time Invariant - Discrete Time Systems

Degree, Semester& Branch: III Semester B.E.ECE A

Course Code & Title: EC8352 Signals and Systems

Name of the Faculty member: Mrs.R.Chitra

Name of the Topic: Convolution Sum

Name of the Innovative Practice: Sorting Strips

Date & Duration: 30.09.2019 & 20 minutes

Description:

Small bits of information are separated into strips so that students can sort the strips into various categories, or organize them into a sequence depending on the topic. This strategy encourages discussion of competing ideas or organizations or order in which a process would take place. In this case, it is often the discussion and sharing of ideas that is the most important outcome of the activity.

Goals (Learning Outcomes):

- The students will be able to analyze Linear Time Invariant Discrete time systems using convolution sum.

Use of appropriate methods:

Justification for choosing the topic using Sorting Strips activity:

A convolution is a mathematical way of combining two signals to form a third signal which is used to analyze the LTI (Linear and Time-Invariant) system or filter using Z transform. Students must know about this topic. But, it is difficult to study for students. It will be useful to analyze digital filters and to analyze the linear discrete system. So, I have chosen this topic to conduct this activity for making students to share their ideas among their peers.

Effective presentation:

Details of the Implementation:

- The goal of this activity is for students to order a set of items, such as steps for solving problems.
- For this, I made entire class into smaller groups (4 to 5 students) and given pieces of papers to each group.
- As one strategy, I given students with a list of procedures for performing convolution operation written on strips of paper that are jumbled
- Students were asked to work together to reconstruct the proper sequence.

Sample images:



Significance of Results:

Assessment of Effectiveness/Success of the Activity:

Usually the students don't attend this topic convolution sum since it is a lengthy procedure. But after conducting this activity they got a clear idea about this topic and most of the students attended this question asked in Internal Assessment Test III.

Reflective Critique:

Benefits

- This activity helps students apply what they have learned through reading or didactic teaching.
- This approach can strengthen students' logical thinking processes and test their mental model of a process.
- In this case, it is often the discussion and sharing of ideas that is the most important outcome of the activity.

Challenges

- I planned to conduct this activity in 15 minutes. But it took around 20 minutes.

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO5	2	3	2	1	1			1		2		2

CO5: The students will be able to interpret the linear time invariant discrete time systems using DTFT and Z transform.

References:

1. <https://blogs.baruch.cuny.edu/saccactivelearning/>
2. <https://teaching.berkeley.edu/active-learning-strategies>
3. <http://www.wlxt.uestc.edu.cn/wlxt/ncourse/dsp/web/kj/Chapter2/2.6.1%20Convolution%20Sum.htm>